Emmanuel Rodriguez

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EDUCATION

St. Mary's University | San Antonio, TX **May 2025**

Bachelor of Science, Computer Engineering

- **Relevant Coursework:** Software Engineering, Object-Oriented Prog and Design, Computer Architecture, Microprocessors, Circuit Analysis/Electronics, Digital Systems Design, Logic Design
- **Honors and Awards:** Gateway Scholarship

SKILLS

Technical /Computer Skills: C/C++, Python, RISC-V, GitHub, HTML, CSS, JavaScript, React, Node.js, VHDL, GDB, Git, Linux, SolidWorks, Jira

TECHNICAL PROJECTS

DermaScope – A Derma-light Skin Analysis System

- Engineered a portable high-resolution skin-imaging device for early detection of potential skin cancer using **Raspberry Pi Pico** and **MicroPython**, enabling **SPI** camera control and **TCP communication** with a companion mobile app.
- Developed a **mobile app** utilizing **Flutter** to analyze captured images using a **TensorFlow model** to assess potential skin cancer risks and securely transmit results to healthcare providers.
- Coordinated with student team members and research mentors from Baylor College of Medicine and UT Health San Antonio to review progress, troubleshoot issues, and refine system design.

RC Drone Car – An Autonomous Navigation System

- Developed a **C-based control system** for manual and autonomous navigation, integrating **Raspberry Pi** remote access via **VNC Viewer**.
- Implemented a **curve correction algorithm** to **optimize trajectory accuracy by ~75%**, securing 2nd place in class track race.

Vision-Based Real Time Tracking System

- Engineered a **C++ vision-based tracking application** for real-time object detection and tracking, computing velocity and trajectory metrics for motion analysis.
- Designed a scalable architecture for storing and processing object data.

Web-app Portfolio – https://e-rodriguez-portfolio.netlify.app/

- Built and deployed **web applications** using **HTML**, **CSS**, **JavaScript**, including a **React-based** mock library website demonstrating **modern UI** and **front-end design.**

EXPERIENCE

JADE Controls – Industrial Automation Engineer | Dallas, TX August 2026 – Present

- Program and troubleshoot PLC control systems using Structured Text and ladder logic, implementing timers, counters, motor controls, interlocks, and sensor-based automation.
- Configure and troubleshoot industrial control systems involving PLC I/O, HMIs, instrumentation, 4–20 mA signals, and industrial communication networks..

TechServ – Engineering Technician | Fort Worth, TX June 2026 – August 2026

- Used proprietary drafting software (CAD) for client projects, including newbuilds, alterations, redevelopments, etc. and work with Engineers to implement design changes into the system.
- Execute changes to designs or sketches and redlines or mark-ups on ongoing projects.

RDZ Quality Homes & Construction – Laborer | San Antonio, TX May 2023 - Present

- Lead a small team in multi-phase construction projects such as **installing electrical wiring and conduit**, assisting with plumbing and structural tasks, and **reading blueprints** to execute work accurately and efficiently using professional-grade equipment.

SAMSAT – R&D Intern | San Antonio, TX January 2025 – May 2025

- Restored functionality by rebuilding a **SCADA** smart grid for a Cyber City museum exhibit using a **Raspberry Pi** and **OpenPLC** to improve visitor engagement.
- Enhanced system architecture by connecting and soldering additional **Raspberry Pi** units, increasing scalability and reliability for future interactive features.

Northern Tool & Equipment – Cashier | San Antonio, TX August 2023 – June 2024

- **Assisted customers** with purchases, product inquiries, and issue resolution while **maintaining front-end operations.**
- Accurately processed transactions and managed daily cash reconciliation, ensuring financial integrity and compliance.

LEADERSHIP

Mariachi Cascabel | San Antonio, TX September 2022 – May 2025

Treasurer, August 2023 – May 2025

Risk-Management Officer, August 2024 – May 2025

- Managed club budget and funds by coordinating financial planning and expenses for organizational activities.
- Assessed risks and ensured compliance with university policies through standardized school training for officer safety.
- Performed in school-wide organization productions and events.

IEEE Student Organization | San Antonio, TX January 2024 – May 2025

Member

- Placed 4th out of 8 teams at the Institute of Electrical and Electronics Engineers (IEEE) Region 5 Greentech Cyber Challenge focused on topics including Cryptography and Reverse Engineering.
- Utilized **Visual Studio Code, Unix / Linux environments**, and leveraged various online resources including dcode.fr and Cyberchef to successfully solve the CTF challenges.